

Assignment 3: CNN Implementation and Evaluation

Einführung in Deep Learning

Due date: 26.06.2026, 09:50

Introduction

In this assignment, you will implement and train your own Convolutional Neural Network (CNN) for image classification. You will work with a dataset that you will create through crowdsourcing, containing images of five clothing objects: T-Shirts, Pants, Socks, Shorts and Long-sleeved. This assignment will guide you through the complete machine learning pipeline: from data analysis to model implementation, training, and evaluation.

Learning Objectives

By completing this assignment, you will:

- Gain hands-on experience in implementing CNNs using PyTorch.
- Gain practical experience in analyzing and preprocessing image dataset
- Implement a CNN architecture from scratch
- Train and evaluate a CNN model on a real-world dataset
- Evaluate model performance using appropriate metrics
- Apply techniques to improve model generalization

Dataset

You will use the crowdsourced dataset you created, which contains images of T-Shirts, Pants, Socks, Shorts and Long-sleeved. Ensure that you have the dataset ready and properly organized for training and evaluation.

Tasks

1. Dataset Analysis

- Load and explore the dataset. Report the number of images per class.
- Visualize sample images from each class.
- Analyze the distribution of image sizes, aspect ratios, and color channels.
- Identify potential challenges in the dataset (class imbalance, image quality issues, etc.).
- Split the dataset into training (80%), validation (10%), and test (10%) sets.

2. Data Preprocessing

- Implement data augmentation techniques (random cropping, flipping, rotation, color jittering) to enhance the training dataset.
- Normalize the images based on the dataset's mean and standard deviation.
- Create PyTorch DataLoader objects for the training, validation, and test sets.
- Visualize some augmented images to verify the preprocessing steps.

3. CNN Implementation

- Implement a CNN architecture from scratch using PyTorch. Your model should include:
 - Multiple convolutional layers with appropriate activation functions and pooling.
 - Fully connected layers leading to the output layer.
 - Dropout layers to prevent overfitting.
- Justify your architectural choices (number of layers, filter sizes, etc.).
- **The total number of parameters in your model should not exceed 800.000**

4. Model Training

- Define an appropriate loss function and optimizer for training.
- Train your CNN on the training set, using the validation set to monitor performance and prevent overfitting.
- Implement early stopping based on validation loss.
- Plot training and validation loss/accuracy curves over epochs.

5. Model Evaluation

- Evaluate your trained model on the test set.
- Report the following metrics:
 - Overall accuracy
 - Class-wise precision, recall, and F1-score
 - Confusion matrix
- Analyze misclassified images and discuss potential reasons for misclassification.

Important Notes:

The use of Weights & Biases (wandb) is mandatory for this assignment. You must log your training runs and upload them.

If you are not in the wandb team of EiDL use this snippet to upload your runs. Be aware you will not be able to see them unless you ask for access. In this case you have to log twice on your own wandb account and in the EiDL team.

Code Snippet:

```
ENTITY = "eidl-thm"
PROJECT = "abgabe2"
run_name = f"Übungsgruppe_Vorname1_Name1_Vorname2_Name2_run_id_changed_parameters_timestamp"

with wandb.init(
    entity=ENTITY,
    project=PROJECT,
    name=run_name,
    ...
) as run:
```

If you are already in the team:

```
PROJECT = Übungsgruppe_Vorname1_Name1_Vorname2_Name2
run_name = f"run_id_changed_parameters_timestamp"
```

Submission Guidelines

Submit a ZIP file containing:

1. Your Python code for data preparation, model implementation, and training.
2. Any additional files necessary to run your code (e.g., trained model weights).
3. A one-pager (PDF, font size 12 pt) summarizing your approach, results, and any challenges you faced during the assignment. This includes:
 - A brief description of your data preprocessing steps (Only if different from the provided code).
 - An overview of your CNN architecture and hyperparameters.
 - Training curves and evaluation metrics.
 - Examples of misclassified images with short analysis.

Be prepared to present your results in the exercise session.

Deadline

The deadline for this assignment **26.06.2026, 09:50** Late submissions will not be accepted unless prior arrangements have been made.